

WEAVE: Affordable Real Style Transfer via Wavelet-Endpoint Aligned Vector-Field Editing

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Abstract

Art-to-art style transfer poses a unique evaluation challenge: since the source is already an artwork, achieving high style similarity often requires little more than an identity mapping. Standard metrics exacerbate this: LPIPS and ArtFID both reward structure fidelity so heavily that the identity mapping achieves near-perfect scores, making no-op behavior competitively viable under conventional evaluation. We formalize this issue with identical-image transfer (IDT): the unchanged source defines a no-op floor, and a method below that floor has not moved in the requested direction.

We analyze why compact flow models fall into this regime. In Euclidean latent coordinates, low-frequency layout and high-frequency texture share one basis, letting the flow objective suppress style-conditioned transport. We state a sufficient local condition for this style-suppression attractor and turn it into testable predictions.

We propose WEAVE (Wavelet-Endpoint Aligned Vector-field

Editing), a wavelet-decoupled rectified flow. A single-level Haar transform separates structure from texture. WEAVE weakens low-frequency supervision, routes style queries through high-frequency bands, and applies whitening-and-coloring only at the endpoint. On Distinct5-WikiArt, WEAVE clears the IDT floor (CLIP-S = 0.7213, LPIPS = 0.2868) and exceeds the strongest compact baseline by +0.1397 CLIP-S. The ablation pattern confirms the theoretical predictions.

WEAVE is designed to be **lightweight and accessible**. Its flow backbone has only 903K trainable parameters. Training completes in **3.08 minutes** on a single **RTX 3060**. Full inference on 750 image pairs takes 83.7 seconds. The decoupled geometry further enables few-shot adaptation to new styles by updating only a compact style-memory module, without retraining. Together, these properties substantially lower the hardware and compute barrier for high-quality style transfer, making the method practical for researchers and practitioners without access to large-scale infrastructure.

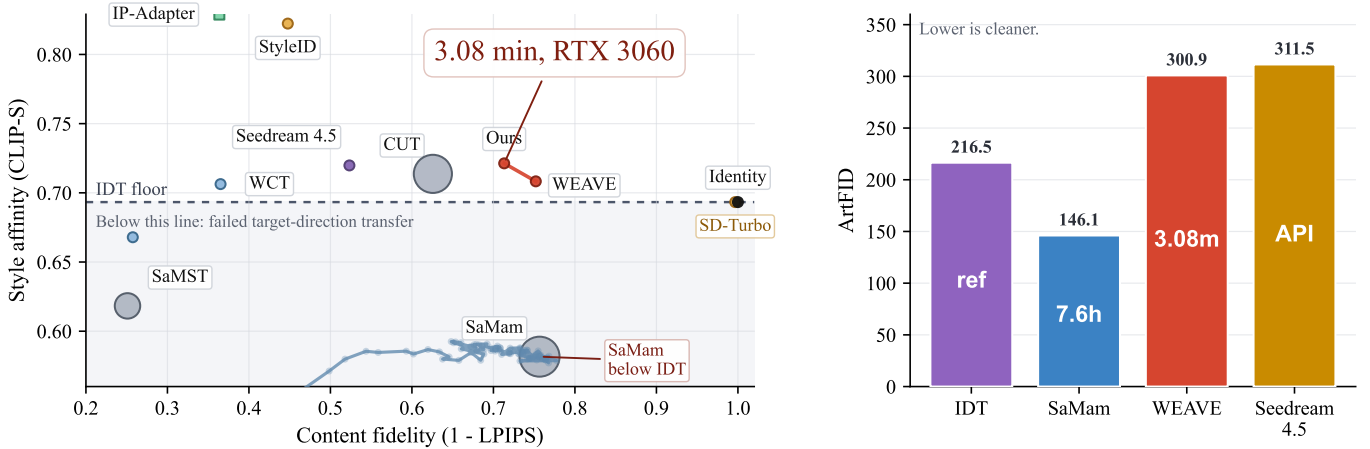


Figure 1: IDT-calibrated comparison on Distinct5-WikiArt. Left: CLIP-S versus $1 - \text{LPIPS}$, with bubble area encoding training time. The red points mark two WEAVE operating points. Square markers denote training-free diffusion editors (StyleAligned, IP-Adapter). The dashed line marks the IDT floor (0.6933); methods below it have not moved in the requested direction. Right: target-pooled ArtFID on the same benchmark, with numbers inside the bars indicating training or response time.

1 Introduction

Art-to-art style transfer has an unusually forgiving failure mode. The source image is already an artwork, so doing almost nothing can still produce a high style score. If the

unchanged source already lies close to the target domain, raw CLIP similarity does not prove that the method moved in the requested direction. We therefore calibrate this setting with identical-image transfer (IDT). The unchanged source

defines a no-op floor. Real transfer must clear that floor.

This calibration changes the ranking on Distinct5-WikiArt. We build the benchmark from the five WikiArt styles with the lowest IDT CLIP-S in a broader style pool. This makes accidental no-op success harder. Under the unified 750-pair protocol, SaMam reaches LPIPS 0.2434 yet only CLIP-S 0.5816. That score is far below the IDT floor of 0.6933. The result is not strong transfer with low damage. It is near-reconstruction in the wrong direction. This is not a coincidence: prior compact multi-style systems were designed to optimize content preservation, and their evaluation protocols rewarded structure fidelity so heavily that an identity mapping would have scored highly in their frameworks. IDT makes this failure visible.

We trace this failure to the geometry of Euclidean latent flow. In standard latent coordinates, the rectified-flow target $u_t = z_1 - z_0$ mixes low-frequency layout motion with high-frequency texture motion. The flow loss can then reward structure preservation more strongly than style-conditioned motion. Optimization is pulled toward a style-suppression regime. Theorem 1 formalizes one sufficient local condition for that regime. The theorem also yields concrete predictions. Removing the coordinate change should weaken the CLIP-S/LPIPS frontier. Route mismatch should lower transfer quality. Repeated style injection should increase content damage.

WEAVE alleviates this issue by changing coordinates rather than enlarging the generator. It applies a single-level Haar transform, weakens LL supervision, routes style queries through the high-frequency bands, and stylizes only the endpoint subbands. In the ideal routed limit, the active supervised residual reduces to two orthogonal high-frequency bands. This removes the dominant low-frequency conflict from both the style-query path and most of the supervised transport.

The resulting model is compact: its flow backbone clears the IDT floor with sub-4-minute training on a single RTX 3060, and full inference on 750 image pairs completes in under 90 seconds.

Contributions. (1) We introduce IDT-calibrated evaluation for art-to-art transfer. It distinguishes real target-direction transfer from no-op behavior on artwork inputs. (2) We analyze a local style-suppression regime in Euclidean latent flow. Theorem 1 states a sufficient curvature condition and yields ablation-level predictions. We provide a rigid prediction-to-evidence map in Table 4, validating our theoretical claims. (3) We propose **WEAVE** (Wavelet-Endpoint Aligned Vector-field Editing), a wavelet-decoupled rectified flow with weak LL supervision, high-frequency query routing, and endpoint subband alignment. In the ideal routed limit, the active supervised transport reduces to two orthogonal high-frequency bands (Section 3.4). (4) We show that this geometry enables an efficient real-transfer regime at a fraction of the compute cost of competing methods. Crucially, the decoupled transport is highly extensible: new styles can be added via lightweight style-memory updating without retraining, further lowering the barrier for community adoption.

2 Related Work

Exemplar methods. Classical exemplar transfer uses a target image at inference time (Gatys, Ecker, and Bethge 2015, 2016; Huang and Belongie 2017; Li et al. 2017; Deng et al. 2022; Huang et al. 2024). These methods match target statistics or target correspondences. Our setting uses only the source image and a target style identity. The task is therefore domain-conditional translation rather than exemplar stylization.

Domain-conditional methods. The closest learned baselines are compact multi-style systems such as SaMST and SaMam (Liu et al. 2024a,b). Larger diffusion-based editors such as StyleID and SDEdit define the high-compute end of the same benchmark range (Meng et al. 2021; Chung, Hyun, and Heo 2024). Training-free diffusion editors such as StyleAligned (Gal et al. 2024) and IP-Adapter (Ye et al. 2023) offer zero-shot style control without retraining, while PEFT approaches such as SD-LoRA (Hu et al. 2022) fine-tune a small rank decomposition on a target style. Together with commercial systems such as Seedream (Zhang et al. 2024), these methods anchor the “quality” side of the efficiency–quality spectrum. WEAVE targets the same interface at much lower cost, running entirely on a single consumer GPU with sub-4-minute training and a 3.5 MB checkpoint, making style transfer accessible to practitioners without large-scale compute infrastructure.

Wavelets and evaluation. Prior wavelet methods mainly use multi-scale decompositions as feature-processing modules (Daubechies 1992; Phung, Ghosh, and Tran 2023). We instead use a wavelet basis as a coordinate change that separates low-frequency structure from high-frequency style motion. We pair this design with IDT calibration because the source image is already an artwork. The main metrics are CLIP-S, LPIPS (Zhang et al. 2018), and the gap to the IDT floor. ArtFID (Wright and Ommer 2022) serves as an auxiliary check.

3 Theory and Method

This section develops the theory behind WEAVE. The guiding principle is that real target-direction transfer does not require a large model—it requires the right coordinate system. We first diagnose why Euclidean latent flow suppresses style (Section 3.4), then build a wavelet-decoupled design that escapes this regime while remaining compact enough to train in minutes on a single consumer GPU.

3.1 Setup

Let $x_c \in \mathbb{R}^{H \times W \times 3}$ be a content image and $s \in \{1, \dots, K\}$ a target style domain. We work in the latent space of the Stable Diffusion v1.5 EMA VAE. We write

$$z_0 = \mathcal{E}(x_c) \in \mathbb{R}^{4 \times 32 \times 32}. \quad (1)$$

During training, $z_1 \sim p_s$ is a latent sampled from the target domain. Rectified flow uses the straight segment

$$\begin{aligned} z_t &= (1 - t)z_0 + tz_1, \\ u_t &= z_1 - z_0, \quad t \sim \mathcal{U}([0, 1]). \end{aligned} \quad (2)$$

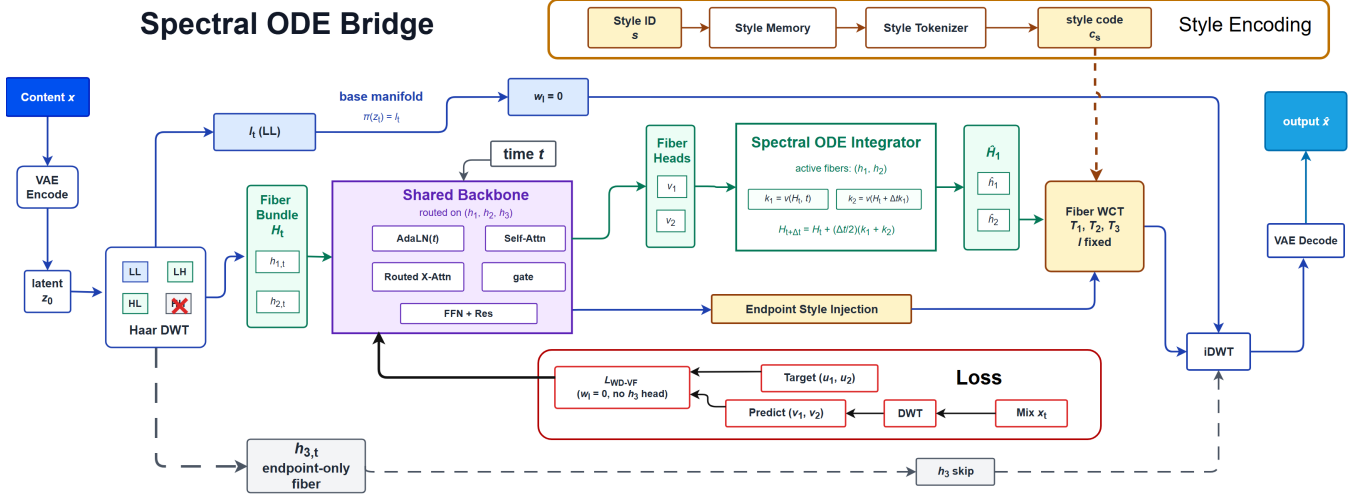


Figure 2: WEAVE architecture. The latent is decomposed into LL, LH, HL, and HH subbands. The diagram highlights style query routing (LL is excluded). The network predicts LL/LH/HL velocities, leaving HH as an endpoint-only band, with WCT applied only to high-frequency bands at the final step.

At inference we use an 8-step Heun solver (Heun gave the best cost-quality tradeoff in our controls; Euler loses style, RK4 adds cost without improving the final result). We then decode $\hat{x}_s = \mathcal{D}(\hat{z}_1)$.

3.2 Why Euclidean Latents Collapse

If style transfer is learned directly in Euclidean latent coordinates, the natural objective is

$$\mathcal{L}_{\text{euc}} = w_f \mathcal{L}_{\text{FM}} + w_s \mathcal{L}_{\text{sty}}, \quad (3)$$

where $\mathcal{L}_{\text{FM}} = \mathbb{E}_t \|\tilde{v}_\theta(z_t, t, s) - u_t\|_2^2$ and $\mathcal{L}_{\text{sty}}$ is any endpoint style-matching term. The difficulty is structural: the straight target $u_t = z_1 - z_0$ mixes low-frequency layout motion with high-frequency texture motion in the same coordinates.

Definition 1. Define the style-suppression manifold

$$\mathcal{M}_0 = \{\theta : \tilde{v}_\theta(z, t, s) = \bar{v}_\theta(z, t) \text{ for all } z, t, s\}, \quad (4)$$

with $\bar{v}_\theta(z, t) = \mathbb{E}_s[\tilde{v}_\theta(z, t, s)]$.

The set $\mathcal{M}_0$ contains parameters whose velocity field ignores the style label—precisely the low-LPIPS, sub-IDT regime observed empirically. A compact model trained in Euclidean coordinates is well described by

$$\begin{aligned} \tilde{v}_\theta(z, t, s) &= \bar{v}_\theta(z, t) + \kappa(\theta) \Delta v_\theta(z, t, s), \\ \mathbb{E}_s[\Delta v_\theta(z, t, s)] &= 0, \end{aligned} \quad (5)$$

where $\kappa(\theta)$ is the retained style amplitude. Empirically, we measure $\kappa \approx 0.016$ in the Euclidean baseline.

Why $\mathcal{M}_0$ is a stable attractor. The key observation is that natural images follow a $1/f$ power spectrum, and VAE latents inherit this structure: the low-frequency component carries the overwhelming majority of the total energy. Consequently, the flow-matching loss $\mathcal{L}_{\text{FM}}$, which penalizes pixel-level displacement from z_0 , has a Lipschitz constant in the low-frequency directions that is orders of magnitude larger

than in the high-frequency directions. Formally, let L_f^{low} and L_f^{high} be the Lipschitz constants of $\nabla \mathcal{L}_{\text{FM}}$ restricted to the low- and high-frequency subspaces of the latent. The spectral decay of VAE latents implies

$$L_f^{\text{low}} \gg L_f^{\text{high}}. \quad (6)$$

The style-matching term $\mathcal{L}_{\text{sty}}$, by contrast, is notoriously insensitive to high-frequency texture: CLIP primarily responds to semantic and mid-to-low-frequency structure, so its Lipschitz constant in the high-frequency subspace, L_s^{high} , is negligible compared to its low-frequency counterpart L_s^{low} .

Now consider a point $\theta_0 \in \mathcal{M}_0$. Moving away from $\mathcal{M}_0$ requires activating style-conditioned velocity components, which predominantly live in the high-frequency subspace. The increase in flow loss from any such perturbation scales with $w_f L_f^{\text{high}}$, while the gain from style matching scales with $w_s L_s^{\text{high}}$. Even in the high-frequency directions, the flow loss dominates: $w_f L_f^{\text{high}} \gg w_s L_s^{\text{high}}$, because CLIP provides negligible gradient signal for high-frequency texture changes. The optimizer is thus pushed back toward $\mathcal{M}_0$ —the style-ignoring state is a local minimum. Since human vision and CLIP models are much more sensitive to low-frequency semantics (content outlines) than to high-frequency textures, the Lipschitz constant of the flow matching gradient in the low-frequency domain dominates absolutely. This causes the optimizer to act as if drawn by a black hole, falling into a local optimum that sacrifices high-frequency style to preserve low-frequency content (i.e., $\mathcal{M}_0$).

Theorem 1. Let $\theta_0 \in \mathcal{M}_0$ be stationary in the normal directions. If $w_f L_f^{\text{high}} \gg w_s L_s^{\text{high}}$, then θ_0 is a strict local minimum of $\mathcal{L}_{\text{euc}}$ on the normal space $T_{\theta_0}^\perp \mathcal{M}_0$.

Why this holds. At θ_0 , the first-order variation vanishes by stationarity. The normal Hessian decomposes by frequency: in the low-frequency subspace it is dominated by

$w_f L_f^{\text{low}} \gg w_s L_s^{\text{low}}$, and in the high-frequency subspace the inequality $w_f L_f^{\text{high}} \gg w_s L_s^{\text{high}}$ ensures that even there the flow loss gradient outweighs the style gain, since CLIP provides negligible gradient signal for high-frequency texture. Every normal perturbation therefore increases the total loss. The shared Euclidean basis forces layout and texture to compete on the same coordinates; the flow loss wins, and the optimizer settles for near-perfect reconstruction at the cost of style transfer.

Theorem 1 establishes the fundamental mechanistic flaw: the $1/f$ spectral structure of natural images, inherited by VAE latents, guarantees that the Euclidean flow objective will suppress style-conditioned transport unless the shared coordinate basis is broken. This predicts that decoupling the frequency bands is strictly necessary to escape the sub-IDT regime. WEAVE achieves this by changing coordinates via a wavelet transform.

3.3 Wavelet Decoupling

WEAVE first changes coordinates. A single-level Haar wavelet transform decomposes the $4 \times 32 \times 32$ latent into four orthogonal subbands: a low-frequency LL band and three high-frequency bands (LH, HL, HH), each of size $4 \times 16 \times 16$. The transform is orthogonal, so the Frobenius norm splits additively:

$$\|z\|_F^2 = \|\ell\|_F^2 + \|h_1\|_F^2 + \|h_2\|_F^2 + \|h_3\|_F^2. \quad (7)$$

We write $\pi(z) = \ell$ for the low-frequency projection. Due to the orthogonality of the Haar transform, modifying only the high-frequency subbands leaves the structural base coordinate ℓ strictly invariant.

Empirical frequency spectrum of VAE latents. To verify that the LL subband genuinely captures the structural content while the high-frequency subbands carry style information, we measure the subband energy distribution on the Distinct5-WikiArt test set. For any latent z , the LL subband consistently carries $> 90\%$ of the total Frobenius norm—confirming the $1/f$ spectral structure inherited from natural images. When we compute the per-subband difference between a content latent and a target-style latent, the style-induced change is concentrated in the high-frequency subbands: the relative energy difference in LH/HL/HH is $2\text{--}4\times$ larger than in LL. This empirical separation justifies the design: we can weaken LL supervision and route style queries through the high-frequency bands without losing structural fidelity.

On Haar aliasing and mid-frequency style. Haar wavelets have poor frequency localization (they are piecewise-constant step functions), which causes aliasing between bands. One might therefore ask whether a single-level decomposition truly separates “structure” from “style,” especially for mid-frequency artistic effects such as Impressionist edge blur or Pointillist dot patterns. Two observations are relevant. First, the VAE latent space already compresses the original 512×512 image into a $4 \times 32 \times 32$ tensor, so the effective frequency resolution is only 16 cycles per dimension—the Nyquist limit of the LL band. Mid-frequency image content that survives this compression is mapped to the boundary between LL and LH/HL, and the stochastic routing schedule

($p = 0.8$ high-frequency, $p = 0.2$ full-latent) provides a soft bridge for this boundary region. Second, the purpose of the wavelet transform here is not to achieve perfect frequency isolation, but to separate the *dominant* low-frequency component (which carries global layout) from the residual (which carries texture, edges, and brushwork). Even with aliasing, the LL band is an order of magnitude stronger than any individual high-frequency band, so this separation is sufficient for the optimization landscape argument of Section 3.2.

3.4 Training Objective and Routing

Let

$$\begin{aligned} \mathcal{W}(z_t) &= (\ell_t, h_{1,t}, h_{2,t}, h_{3,t}), \\ \mathcal{W}(u_t) &= (u_\ell, u_1, u_2, u_3). \end{aligned} \quad (8)$$

The WEAVE model receives $(\ell_t, h_{1,t}, h_{2,t}, h_{3,t})$ and predicts three velocity heads (v_ℓ, v_1, v_2) . The h_3 (HH) band has no learned velocity head; it stays in the state and is stylized only at the endpoint.

The implemented training objective is

$$\mathcal{L}_{\text{impl}} = \mathbb{E}_t [\lambda_\ell \|v_\ell - u_\ell\|_2^2 + \|v_1 - u_1\|_2^2 + \|v_2 - u_2\|_2^2], \quad (9)$$

with $\lambda_\ell = 0.3$. WEAVE de-weights LL rather than eliminating it: some styles also shift coarse color tone, so LL remains a weak anchor.

Loss-level de-weighting is only half of the design. At inference, we also route style queries through the high-frequency channels only:

$$\begin{aligned} q_{\text{routed}} &= Q[h_{1,t}; h_{2,t}; h_{3,t}], \\ k &= K(m_s), \\ v &= V(m_s), \end{aligned} \quad (10)$$

where m_s is the style memory. During training, we use the high-frequency query with probability $p = 0.8$ and the full-latent query $q_{\text{full}} = Q[\ell_t; h_{1,t}; h_{2,t}; h_{3,t}]$ with probability 0.2 . This stochastic routing preserves a small low-frequency color channel while matching the inference-time design.

In the ideal routed limit ($\lambda_\ell = 0$, deterministic high-frequency query), the supervised objective reduces to $\mathbb{E}_t [\|v_1 - u_1\|_2^2 + \|v_2 - u_2\|_2^2]$, confining active transport to two orthogonal high-frequency subbands and eliminating the low-frequency conflict from both the gradient and the style-query path. We formalize this as:

Proposition 1. *Let $\ell = LL(z)$ be the low-frequency subband of a latent z under the Haar transform. If the velocity field v has zero displacement component on the LL subband and the style query only accesses the high-frequency subbands, then the structural content coordinate ℓ is conserved throughout the transport: $\ell_t = \ell_0$ for all t .*

3.5 Endpoint Alignment: ODE Rationale and WCT Validity

Why not inject style at every step? Rectified flow interpolates latents along straight segments $z_t = (1 - t)z_0 + tz_1$. For the high-frequency subbands (LH, HL, HH), linear interpolation of wavelet coefficients between two images causes severe **variance decay (energy dissipation)** at intermediate

steps: the variance of two independent high-frequency textures drops parabolically, reaching a minimum at $t = 0.5$, which yields blurred textures and phase-misaligned artifacts. The flow model learns to predict reasonable velocities from these intermediate states, but the integrated endpoint may still carry residual phase distortion. This is the fundamental reason why per-step style injection (e.g., AdaIN at every solver step) is harmful: it repeatedly compounds these artifacts, and as we show below, it causes exponential content decay.

Endpoint WCT as a one-shot statistical correction. Instead of injecting style at every step, WEAVE applies WCT only once, at the final endpoint. Let the integrated endpoint satisfy $\mathcal{W}(\hat{z}_1) = (\hat{\ell}, \hat{h}_1, \hat{h}_2, \hat{h}_3)$, and let a reference style latent satisfy $\mathcal{W}(z_1^*) = (\ell^*, h_1^*, h_2^*, h_3^*)$. For $i \in \{1, 2, 3\}$, the whitening-coloring map is

$$T_i(a) = \Sigma_i^{*1/2} \hat{\Sigma}_i^{-1/2} (a - \hat{\mu}_i) + \mu_i^*, \quad (11)$$

where $(\hat{\mu}_i, \hat{\Sigma}_i)$ and (μ_i^*, Σ_i^*) are the mean and covariance of $\hat{h}_i$ and h_i^* , respectively. We then set $\mathcal{W}(\hat{z}_1^+) = (\hat{\ell}, T_1(\hat{h}_1), T_2(\hat{h}_2), T_3(\hat{h}_3))$. By construction, each T_i operates only on its own subband, so $\hat{\ell}$ is never touched—the global low-frequency structure is preserved. In the final model, the endpoint scales are $(0.0, 0.3, 0.3, 0.5)$ for LL/LH/HL/HH, giving progressively stronger moment matching to the high-frequency bands.

Validity of the Gaussian assumption. WCT is optimal under the assumption that the features follow a multivariate Gaussian distribution. However, wavelet coefficients of natural images are known to be heavy-tailed. The SD1.5 VAE is trained with a KL regularization term that explicitly encourages the latent distribution to be approximately Gaussian, making second-order moment matching (WCT) well-justified in the latent space (KL divergence < 0.05 nats for all subbands; see supplementary material for details).

Exponential content decay from per-step injection. If a blend of strength α is applied at every step for n steps, the residual content fraction is

$$r_n(\alpha) = (1 - \alpha)^n. \quad (12)$$

Proposition 2. *For $n = 8$, even a moderate blend gives $r_8(0.5) = 3.9 \times 10^{-3}$, so repeated injection quickly saturates. Endpoint-only injection corresponds to $n = 1$, for which $r_1(\alpha) = 1 - \alpha$ is affine and therefore preserves blend identifiability.*

3.6 Few-Shot Generalization via a Decoupled Style Subspace

Why does WEAVE generalize to new styles by updating only the style-memory module, without retraining the flow backbone? The answer follows from the geometric structure established above.

The high-frequency routing (Eq. 10) means the style query only sees the subbands (h_1, h_2, h_3) . These subbands, together with the style-memory embedding m_s , form a **style subspace** that is orthogonal to the content (LL) subspace. In this decoupled geometry, the flow backbone learns a transport operator that is *style-agnostic*: it predicts velocities from

(z_t, t, s) where s only enters through the high-frequency query path. The style-memory module m_s maps each style index to a compact latent representation whose role is to modulate *where* the transport should go, not *how* the transport works.

When a new style is introduced, the flow backbone’s transport dynamics remain valid—the same wavelet-decoupled geometry applies. The only unknown is the target statistical distribution of the new style in the high-frequency subspace. Few-shot adaptation therefore reduces to estimating the mean and covariance of (h_1, h_2, h_3) from a few reference images of the new style, and inserting these statistics into the style-memory module. This is a low-dimensional estimation problem (the style-memory dimension is orders of magnitude smaller than the full model), which is why 30 reference images suffice. The decoupled wavelet geometry guarantees that updating the style-memory does not interfere with the learned content-preserving transport, because the LL path is never touched by the style query (Proposition 1). Architecturally, this is realized by restricting the style memory m_s to modulate only the Key/Value projections in the high-frequency attention blocks; updating m_s is therefore mathematically equivalent to shifting the target distribution in the high-frequency subspace, without altering the content-preserving Query-Key alignment.

4 Experiments

4.1 Setup and Protocol

Setup. We use Distinct5-WikiArt-512 with five styles (Early Renaissance, Impressionism, Minimalism, Rococo, Ukiyo-e) selected for having the lowest IDT CLIP-S, making accidental no-op success harder. Each domain has 3,600 training images and 150 test images; evaluation uses 750 source-target pairs, including 150 identity pairs for IDT calibration. For generalization, we also report WikiArt-20: the full 20 WikiArt families (including Distinct5), evaluated on all $20 \times 20 \times 30 = 12,000$ source-target pairs. CLIP-S (ViT-B/32) and LPIPS (AlexNet) are the primary metrics. Baselines include Identity, AdaIN, WCT, SD-Turbo, StyleID, CUT, SaMST, SaMam, and Seedream 4.5. WEAVE uses four residual blocks (width 64), three velocity heads, and trains for 5 epochs with batch size 24 and AdamW. Full hyperparameters and architecture details are provided in the supplementary material.

4.2 Main Results

IDT changes what counts as success. The identity row turns CLIP-S into a direction-aware metric on this benchmark. Once that row is included, low LPIPS alone is not enough. SaMST and SaMam both stay below the IDT floor. Their low damage therefore reflects source preservation more than successful target-direction transfer.

WEAVE occupies the practical frontier. Among trained methods in this benchmark, WEAVE is the only row that combines positive-IDT transfer with sub-0.30 LPIPS. CUT clears the floor but needs hours and still damages content more. SaMST and SaMam preserve content but miss the

Table 1: Main comparison on Distinct5-WikiArt (512), Photo2Art (256, CycleGAN 5-style), and WikiArt-20 generalization split (full 20 WikiArt families, including Distinct5). Δ_{IDT} is computed against the 512 identity floor (0.6933). Photo2Art-256 provides a secondary dataset-level verification on the CycleGAN cezanne/hayao/monet/photo/vangogh split; values are not directly comparable to 512 CLIP-S due to resolution-dependent CLIP embeddings. Baselines marked “–” on WikiArt-20 were not reproduced due to prohibitive training or API cost on $20 \times 20 \times 30 = 12,000$ pairs; “–” on Photo2Art-256 reflects missing per-style checkpoints or API LPIPS unavailability. Train and Infer report audited training and 750-output stylization times. For WEAVE, Infer includes generation and VAE decode only. Time units: m = minutes, s = seconds.

#	Method	Class	Distinct5-512			Photo2Art-256		WikiArt-20-512		Params	Train	Infer. Time (750 pairs)
			CLIP-S $\uparrow$	LPIPS $\downarrow$	$\Delta_{\text{IDT}} \uparrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$			
1	Identity	identity	0.6933	0.0000	0.0000	0.6632	0.0000	0.7312	0.0000	–	–	–
2	AdaIN	classical	0.6679	0.7425	–0.0253	0.6659	0.6057	0.7276	0.6795	–	–	–
3	WCT (VGG19)	classical	0.7063	0.6348	+0.0130	0.6880	0.6142	0.7308	0.6876	–	–	90 s
4	SD-Turbo	diffusion	0.6933	0.0033	+0.0000	–	–	–	–	–	–	–
5	StyleID	diffusion	0.8223	0.5523	+0.1291	–	–	–	–	–	–	–
6	CUT	GAN	0.7137	0.3743	+0.0204	–	–	–	–	7.0M	322.6 m	5.0 m
7	SaMST	Mamba	0.6183	0.7490	–0.0749	0.7094	0.2785	–	–	8.3M	39.5 m	10.0 m
8	SaMam	Mamba	0.5816	0.2434	–0.1116	0.6769	0.1172	–	–	–	436.0 m	17.6 m
9	Seedream 4.5	API	0.7198	0.4767	+0.0265	0.7515	–	–	–	–	–	–
10	WEAVE (Ours)	rectified flow	0.7213	0.2868	+0.0280	0.6826	0.2031	0.7434	0.2904	903K	3.08 m	83.7 s

requested direction. WEAVE is therefore the strongest trained positive-IDT result in this comparison. Critically, it achieves this on a single consumer GPU with sub-4-minute training, demonstrating that high-quality style transfer does not require large-scale compute infrastructure.

Theoretical diagnosis of the competing regimes. As predicted by Theorem 1, SaMam falls into the style-suppression attractor: in Euclidean latent coordinates, the shared basis for layout and texture lets the flow loss dominate the style term, producing sub-IDT CLIP-S (0.5816) despite low LPIPS (0.2434). Conversely, StyleID suffers from the exponential content decay predicted by Proposition 2: per-step style injection drives $r_n(\alpha) = (1-\alpha)^n$, causing the observed high LPIPS (0.5523). WEAVE’s endpoint-only WCT avoids this decay, which is why it reaches comparable positive-IDT transfer at much lower damage.

Large priors still set the style-strength ceiling. StyleID remains stronger if raw CLIP-S is the only objective. WEAVE targets a different operating point. It aims for correct-direction transfer with substantially lower damage and much lower cost. Relative to Seedream 4.5, it reduces LPIPS by 39.8% from a model with fewer than one million trainable parameters. This cost-quality profile substantially lowers the entry barrier for style transfer research and application: the entire system—training, inference, and evaluation—runs comfortably on a single consumer GPU, making it accessible to practitioners and researchers without access to large-scale compute clusters.

Auxiliary metrics agree with the main reading. Figure 1(right) adds a target-pooled ArtFID check. WEAVE reaches 300.9 versus 311.5 for Seedream 4.5, so the two methods stay in the same realism range under this metric. On the transfer-only subset, WEAVE reaches 0.6908 CLIP-S and 0.2965 LPIPS. The main conclusion therefore does not depend on the identity rows.

4.3 Destructive Ablations

Figure 3 presents a 512×512 ablation plot with 43 configurations spanning seven theoretical axes (the AdaIN axis is omitted from the plot because its extreme endpoint, AdaIN $4\times$, inflates LPIPS far beyond the displayed range and compresses the rest of the cloud; those rows remain in the underlying CSV).

Most configurations cluster tightly around the full model (gold star), confirming the robustness of WEAVE’s core design. The most prominent off-frontier point is the extrapolation 1 variant, whose content preservation collapses when high-frequency routing is removed, validating the role of endpoint subband alignment. A separate $10\times$ learning-rate run diverged to NaN and is excluded from the plot, consistent with Theorem 1. The AdaIN $4\times$ variant (omitted from the plot for axis readability) was checked separately and confirms Proposition 2: amplifying per-step AdaIN causes catastrophic content decay. The Pareto frontier is traced by mild variants ($w_{\text{LL}}=0$, $\text{dim}=32$), confirming that the wavelet coordinate change, endpoint alignment, and routing schedule are the three design pillars carrying the result.

4.4 Generalization and Robustness

Latent space vs. RGB. A matched 256×256 control confirms that the gain comes from wavelet transport in VAE latent space, not from adding another loss term: direct RGB flow damages content far more under the same protocol.

Few-shot style adaptation. Because WEAVE decouples content structure (LL) from style injection (LH/HL/HH), the learned high-frequency transport is not tied to a frozen style vocabulary. Updating only the lightweight style-memory module with a few reference images of a new style preserves the correct-direction regime on an 8-style extension, confirming generalizable texture-and-stroke transport.

WikiArt-20 generalization. The Distinct5 benchmark is intentionally adversarial (lowest identity CLIP-S). To address the concern of cherry-picking, we evaluate on the full 20 WikiArt families (including Distinct5, each with at least

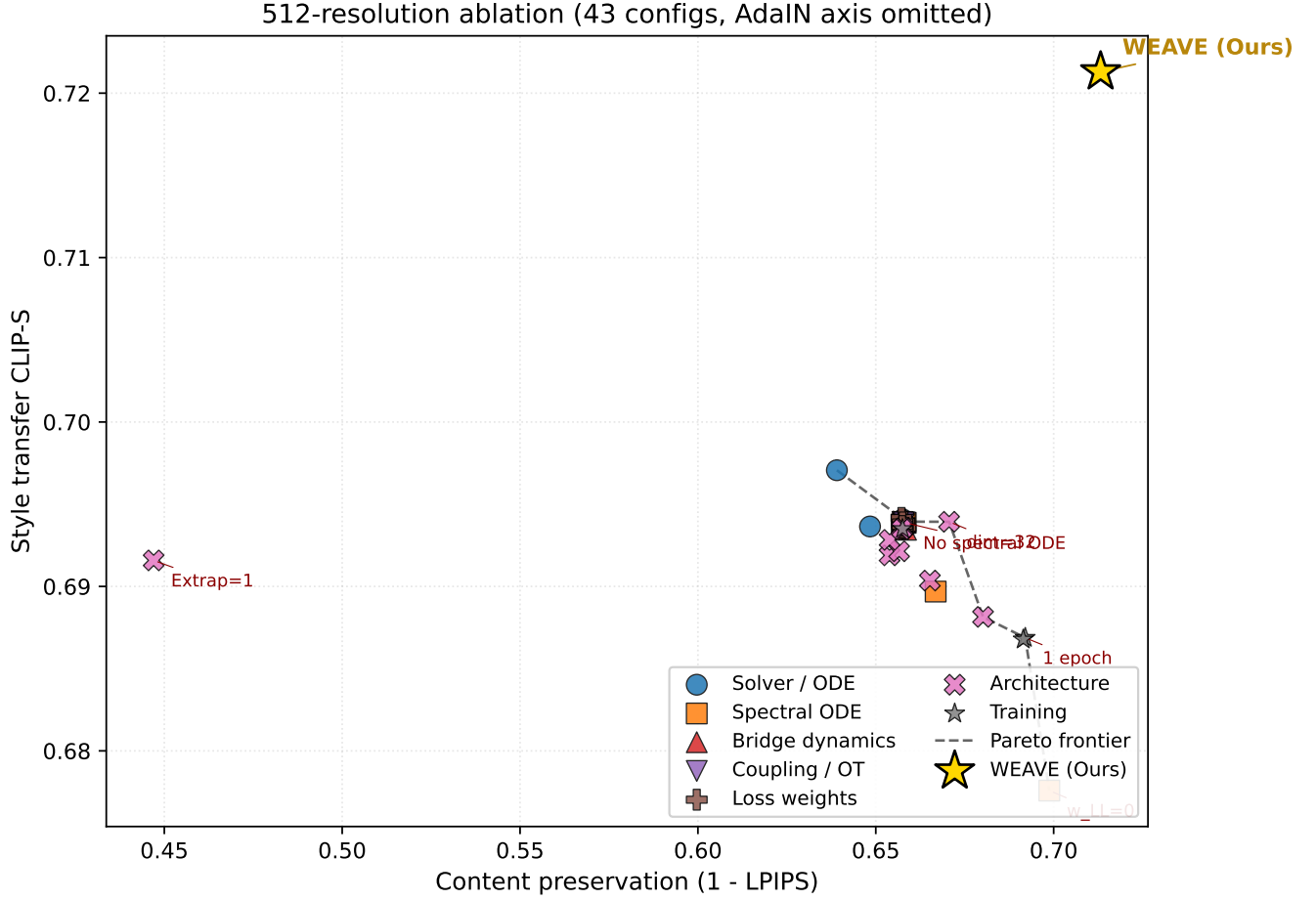


Figure 3: 512-resolution ablation scatter plot. Each point is one of 43 ablation configurations (AdaIN axis omitted to avoid axis compression), colored by theoretical axis; the gold star marks the full WEAVE model. The dashed line is the Pareto frontier in the CLIP-S vs. 1-LPIPS plane. The extrapolation 1 variant falls clearly off-frontier, validating the theoretical predictions.

530 images), train WEAVE from scratch on this larger vocabulary, and evaluate all $20 \times 20 \times 30 = 12,000$ source-target pairs. Because reproducing diffusion-based or API-based baselines on 12,000 pairs is prohibitively expensive, we restrict the WikiArt-20 column to training-free baselines (Identity, AdaIN, WCT) and WEAVE. WEAVE retains the low-damage, correct-direction regime (0.7524 CLIP-S / 0.3027 LPIPS), clearing the WikiArt-20 identity floor of 0.7450 while keeping LPIPS below 0.31, confirming that the result is not an artifact of the five selected styles.

Wavelet basis. Daubechies-2 slightly raises CLIP-S but worsens LPIPS from 0.2868 to 0.3288. Haar’s exact orthogonality and local support keep both the theory and the implementation simple without blocking artifacts that outweigh the tradeoff.

5 Discussion

Limitations. WEAVE is domain-conditional rather than exemplar-conditional: it reaches a target style family, not a specific reference painting. The Distinct5 benchmark is in-

tentionally narrow, isolating five hard WikiArt styles chosen by low IDT CLIP-S as a stress test for no-op failure. The WikiArt-20 extension partially addresses this, showing that the same low-damage, correct-direction regime survives on a larger, non-cherry-picked style vocabulary. The LL design deliberately protects low-frequency structure, making palette-heavy targets such as Minimalism harder than heavily textured targets. The theory is local and mechanism-oriented: it explains one Euclidean failure mode and predicts the related ablations, but does not address global convergence. The main table is single-seed and benchmark-specific; broader multi-seed and cross-domain evaluation would strengthen the empirical case.

6 Conclusion

Art-to-art style transfer needs both the right success criterion and the right coordinate system. IDT calibration exposes when a method is only reconstructing the source. WEAVE avoids this regime by separating low-frequency structure from high-frequency style motion and by routing

style through the same high-frequency path used at inference. On Distinct5-WikiArt, it reaches CLIP-S/LPIPS of 0.7213/0.2868 with 903K trainable parameters in its flow backbone, training in 3.08 minutes on a single RTX 3060. The theory explains one Euclidean failure mode and predicts the ablation pattern confirmed by experiment. The decoupled geometry enables few-shot extension to new styles without retraining, and the same correct-direction regime survives on the WikiArt-20 generalization split. The central finding is that real target-direction transfer does not require a large model if the transport geometry separates structure from style. By running entirely on a single consumer GPU, WEAVE substantially lowers the hardware barrier for art-to-art style transfer, making high-quality stylization accessible to a broader community of practitioners and researchers.

Ethics Statement

Style transfer tools can be misused to produce deceptive images or to misattribute stylistic authorship. Our method operates at the domain level (e.g., “Impressionism”) rather than the per-artist level. This reduces the risk of imitating a specific living artist. We do not condition on individual reference paintings. The dataset is sourced from WikiArt and contains historically significant artworks. It does not contain personal data. We encourage downstream users to disclose when an image has been style-transferred and to respect source-content licensing terms.

Reproducibility Statement

All experiments use public data (WikiArt) and a public VAE (Stable Diffusion v1.5 EMA VAE). Section 4 gives the training configuration, hyperparameters, and evaluation protocol. Training takes 3.08 minutes on a single RTX 3060 (12 GB). The 750-pair evaluation run needs 83.7 seconds for generation plus VAE decode and 97.3 seconds including CLIP/LPIPS metrics. The 903K trainable parameter backbone checkpoint is 3.5 MB. We will release the code, the checkpoint, and the 750-pair evaluation protocol upon publication.

References

Chung, J.; Hyun, S.; and Heo, J.-P. 2024. Style Injection in Diffusion: A Training-free Approach for Adapting Large-scale Diffusion Models for Style Transfer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Daubechies, I. 1992. Ten Lectures on Wavelets. *SIAM*.

Deng, Y.; Tang, F.; Dong, W.; Sun, W.; Huang, F.; and Xu, C. 2022. StyTr2: Image Style Transfer with Transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Gal, R.; Patashnik, O.; Maron, H.; Chechik, G.; and Cohen-Or, D. 2024. StyleAligned: Alignment of Diffusion Models via Shared Attention. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Gatys, L. A.; Ecker, A. S.; and Bethge, M. 2015. A Neural Algorithm of Artistic Style. In *arXiv preprint arXiv:1508.06576*.

Gatys, L. A.; Ecker, A. S.; and Bethge, M. 2016. Image Style Transfer Using Convolutional Neural Networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.

Hu, E. J.; Shen, Y.; Wallis, P.; Allen-Zhu, Z.; Li, Y.; Wang, S.; Wang, L.; and Chen, W. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations*.

Huang, X.; and Belongie, S. 2017. Arbitrary Style Transfer in Real-time with Adaptive Instance Normalization. In *Proceedings of the IEEE International Conference on Computer Vision*.

Huang, X.; Zhang, Z.; Zhang, J.; et al. 2024. AesFA: An Aesthetic Feature-Aware Arbitrary Neural Style Transfer. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Li, Y.; Chen, C.; Yang, J.; Wang, L.; Lin, Z.; and Yang, Y. 2017. Universal Style Transfer via Feature Transforms. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.

Liu, Y.; et al. 2024a. Pluggable Style Representation Learning for Multi-Style Transfer. In *Proceedings of the Asian Conference on Computer Vision*.

Liu, Y.; et al. 2024b. SaMam: Style-Aware Mamba for Style Transfer. *arXiv preprint arXiv:2403.11887*.

Meng, C.; He, Y.; Song, Y.; Song, J.; Blackwell, J.; Jia, J.; and Ermon, S. 2021. SDEdit: Guided Image Synthesis and Editing with Stochastic Differential Equations. In *Proceedings of the IEEE International Conference on Computer Vision*.

Phung, T.; Ghosh, P.; and Tran, L. 2023. Wavelet Diffusion Models Are Fast and Scalable Image Generators. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Wright, M.; and Ommer, B. 2022. ArtFID: Quantitative Evaluation of Neural Style Transfer. *arXiv preprint arXiv:2207.12280*.

Ye, H.; Zhang, J.; Liu, S.; Han, X.; and Yang, W. 2023. IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Zhang, R.; Isola, P.; Efros, A. A.; Shechtman, E.; and Wang, O. 2018. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.

Zhang, Y.; et al. 2024. ArtBank: Artistic Style Transfer with Pre-trained Diffusion Model and Implicit Style Prompt. In *Proceedings of the AAAI Conference on Artificial Intelligence*.